Week 12: 16/11/2026 – 20/11/2026

Subject: Mathematics Course 3 **Unit Title: Exploring Transformations and Linear Relationships** **Grade : 8**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative/تكويني Summative /ختامي Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: Similar triangles-Part 1	I can recognize similar triangles by comparing their corresponding angles and side lengths.	1.1 Flexible display of information (size, color, font, layout). 2.3 Support decoding of text and notation using tools, audio, and multiple representations. 8.1 Make goals clear, visible, and supported by short-term objectives.	**MAGMA MATH:** **Very Important: Students are requested to complete the assignments on Magma Math on weekly basis.** Girls 'campus: Students will check their materials on Files section on Teams.
Lesson 2: Similar triangles-Part 2	I can use the properties of similar triangles to find missing side lengths and solve problems		
Lesson 3: 2D shapes	I can find the area of 2D shapes.		
Lesson 4: 3D shapes	I can find the volume of 3D shapes.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	**Make sure your devices are fully charged.** **Bring your pencil case/calculator with you.** **Complete your extra practice on Magma Math App.** **GRASPS TASK DUE ON NOVEMBER.20.** **Grade 8B will have an official assignment to be submitted weekly according to the teacher's announcement on Toddle.** **Homework: This week, the students need to complete the assignment about area of 2D shapes scheduled on Magma Math.**		

Technology Integration دمج التكنولوجيا	Resources shared on Teams
Cross-curricular links الروابط بين المواد الدراسية	Science Connection: Students explore how slope relates to the steepness of inclined planes and how it affects the force needed to move objects up a ramp. IT Connection: Students use GeoGebra or Desmos to construct similar triangles on a straight line and investigate how the rise/run ratio remains constant.
External Assessment Integration دمج التقييمات الخارجية	All content is covered in MAP
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	🧠 **Wellbeing** • Confidence in solving real-world measurement problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussion. • Self-awareness by checking the reasonableness of solutions. 🌎 **IB ATL Skills** **Thinking:** solving problems using scale factors and proportional reasoning. **Communication:** explaining mathematical methods and solutions clearly. **Research:** interpreting scale drawings and real-world models. **Self-Management:** organising calculations accurately and efficiently. **Social:** discussing strategies and comparing solutions with others
Standards المعايير	8.F.A.1: Understand that a function is a rule that assigns each input exactly one output. 8.F.B.4: Construct a function to model a linear relationship between two quantities and interpret the rate of change and initial value. MP3: Construct viable arguments and critique the reasoning of others. MP7: Look for and make use of structure.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How does the accuracy of a scale factor affect the final size and reliability of a real-world design? How can slope help us understand and compare rates of change in real-life situations? How can a graph help us identify and understand the unit rate in a proportional relationship? How can unit rates help us compare proportional relationships represented in different ways?

Success Criteria معايير النجاح (Measurable)	Lesson 1: 🟦 All students must be able to identify corresponding angles and sides in two triangles and recognize when triangles are similar using angle relationships. 🔵 Most students should be able to use the AA Similarity Criterion to determine whether two triangles are similar and explain my reasoning. 🔺 Some students could be able to apply triangle similarity to solve problems involving unknown angles and explore how corresponding sides are related. Lesson 2: 🟦 All students must be able to identify corresponding sides of similar triangles and set up a proportion to find a missing side length. 🔵 Most students should be able to solve multi-step problems involving similar triangles using proportions and justify the steps. 🔺 Some students could be able to apply triangle similarity to solve real-world and challenging geometric problems, including indirect measurement, and explain my solution. Lesson 3: 🟦 All students must be able to find the area of 2D shapes. 🔵 Most students should be able to find the area of compound 2D shapes 🔺 Some students could be able to solve word problems involving area of 2D shapes. Lesson 4: 🟦 All students must be able to find the volume of 3D shapes. 🔵 Most students should be able to find the volume of compound 3D shapes 🔺 Some students could be able to solve word problems involving volume of 3D shapes.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Respect الاحترام Heritage التراث
	Students connect slope and proportional relationships to **UAE roads, transport routes, and infrastructure**, strengthening their understanding of how mathematics supports their community. Students demonstrate respect by **collaborating, sharing mathematical strategies, and appreciating different approaches** when solving UAE real-world problems. Students explore proportional relationships through **UAE heritage contexts**, such as traditional architecture, falaj irrigation systems, and historical trade routes.